M. Sc. Semester IV, Mid-semester Exam -2025: CSM 402: Introduction to Data Science

Full Marks: 20, Duration: 1 hour

1. Answer any four questions from the following: **2 marks x 4**

a) Find $P(X \leq 1)$ and $P(0.5 \leq X \leq 1)$ Given CDF $F(x) = 0;\ x < 0$

$$= x^2/4;\ 0 \leq x < 2;$$

$$= 1;\ 2 \leq x$$

b) Are the vectors [2 1 -2] and [3 1 3] orthogonal?

c) Give an example of basis vectors that will always span R^4 .

d) If r eigenvalues of an n x n matrix are 0 what can you say about the rank, nullity and column space of the matrix?

e) Define Covariance and Correlation and explain their significance.

2. Answer any three questions from the following: **4 marks x 3**

a) Find the nullity of the below matrix. Are null space vectors linearly independent?

1	2	3
2	4	5
3	6	7

b) Comment on "Obtaining linearly independent vectors from a data matrix may save storage".

c) Show that eigenvalues and eigenvectors of covariance matrix of a given data gives the principal components in PCA.

d) For customers purchasing a refrigerator at a certain appliance store, let A be the event that the refrigerator was manufactured in the U.S., B be the event that the refrigerator had an icemaker, and C be the event that the customer purchased an extended warranty. Relevant probabilities are $P(A) = .75$, $P(B \mid A) = .9$, $P(B \mid A') = .8$, $P(C \mid A \cap B) = .8$, $P(C \mid A \cap B') = .6$, $P(C \mid A' \cap B) = .7$, $P(C \mid A' \cap B') = .3$. i) Construct a tree diagram consisting of first-, second-,and third-generation branches, and place an event label and appropriate probability next to each branch. ii) Compute $P(A \cap B \cap C)$, iii) Compute $P(B \cap C)$ iv) Compute $P(C)$.

M. Sc. Semester IV, Mid-semester Exam 2: CSM 402: Introduction to Data Science

Full Marks: 20, Duration: 1 hour

1. **Answer any *four* questions from the following:** **2 marks x 4**

a) Define unbiased estimator. Give an example of an unbiased estimator.

b) Define p-value. What is the significance of p-value.

c) Assuming porosity of certain material follows a normal distribution with standard deviation of 0.75. Compute a 95% CI for the true average porosity of the material if the average porosity for 20 specimens is 4.85.

d) Define the one-sample CI for μ considering t distribution.

e) Define Null and alternative hypothesis and explain their significance.

f) What are the principal components in principal component analysis?

2. **Answer any *three* questions from the following:** **4 marks x 3**

a) Given $n=14$, $\sum x_i=517$, $\sum y_i=346$, $\sum x_i^2=39.095$, $\sum y_i^2=17.454$; $\sum x_i y_i=25.825$, give simple linear regression model. Define RSS, TSS and coefficient of determination r^2. How they are interpreted to determine the quality of model built

b) Derive the coefficients β_i in Multiple Linear Regression model.

c) Justify how the process of principal component analysis achieves its objective.

d) Define Singular Value decomposition (SVD). Give the steps of obtaining the SVD with justification.
